

Explain It Like You Built It (Week 6 Assignment)

TOPIC: Glassmorphic UI & Ambient CSS Lighting | AUTHOR: Abdul Sami | TRACK: General AI Fluency

1. The Concept I Explored: Glassmorphic UI & Ambient Spheres

When I built my portfolio (<https://abdulsamiuthwal-portfolio.vercel.app/>), I wanted it to feel modern and premium. I kept seeing frosted-glass cards with glowing purple-and-emerald lights floating behind them. At first, it looked like complex 3D graphic magic, but I had AI tutor me on how standard CSS handles this, and now I genuinely own how it works.

2. The Glowing Spheres (Fake 3D Lighting)

Instead of loading heavy 3D image files, I created floating colored HTML divs with a heavy blur filter:

filter: blur(120px); border-radius: 50%;

Plain Words Analogy: Think of putting a bright flashlight inside a frosted glass bottle. The 120px blur softens the hard edges of the circle into a smooth, ambient glow that bleeds into the dark background.

3. The Glassmorphic Cards (The Frosted Glass Effect)

```
/* Glassmorphism CSS in styles.css */
background: rgba(17, 24, 39, 0.65);
backdrop-filter: blur(16px);
border: 1px solid rgba(255, 255, 255, 0.1);
```

- **rgba(17, 24, 39, 0.65):** Gives the card a dark grey color with 35% transparency.
- **backdrop-filter: blur(16px):** Tells the browser to blur whatever sits *behind* the card (the glowing spheres), creating a real frosted glass look.
- **border: 1px solid rgba(255,255,255,0.1):** Adds a razor-thin white border to mimic light reflecting off sharp glass edges.

4. Key Takeaway & Learning

Understanding this transformed how I view web UI. I don't need heavy graphics libraries or pre-rendered images. By combining hardware-accelerated CSS filters (blur), semi-transparent colors (rgba), and backdrop blurring, the browser renders dynamic, responsive depth in real-time with zero performance lag.